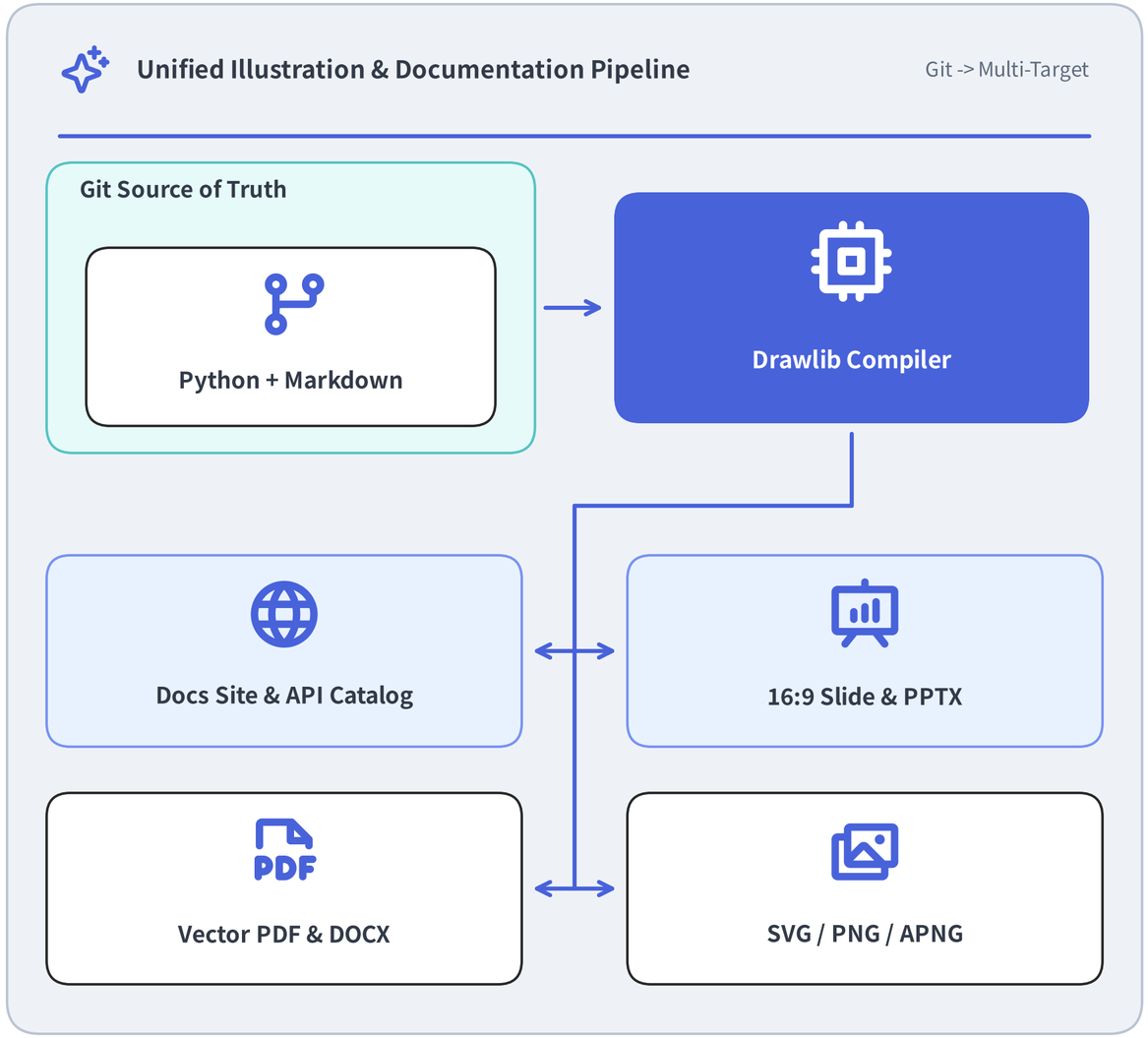
Drawlib

Illustration & Documentation as Code

Pure-Python declarative visuals and multi-target documentation compiler built for software engineers and AI coding agents.

Official Showcase Deck · v0.3.5 Ready



Drawlib at a Glance

4,500+

Vector Icons

Across 34 official & neutral suites

38+

High-Level Visuals

18 SmartArts, 9 Charts,
6 Diagrams, 5 Graphs

5

Project Archetypes

site, catalog, doc, slide, images

100%

Pure Python & Git

Zero GUI lock-in, AI self-verifiable

Four-Layer Architecture Stack

From AI CLI & Primitives to Multi-Target Publishing

L4 · Layer 4 · Multi-Target Compiler

HTML · Catalog · Slide · PDF · DOCX · PPTX

L3 · Layer 3 · Declarative Visual Engines

SmartArts · Charts · Diagrams · Graphs · Anim

L2 · Layer 2 · Primitives, 34 Icon Suites & Semantic Styles

43 Shapes · 4,500+ Vector Icons · Theme Presets

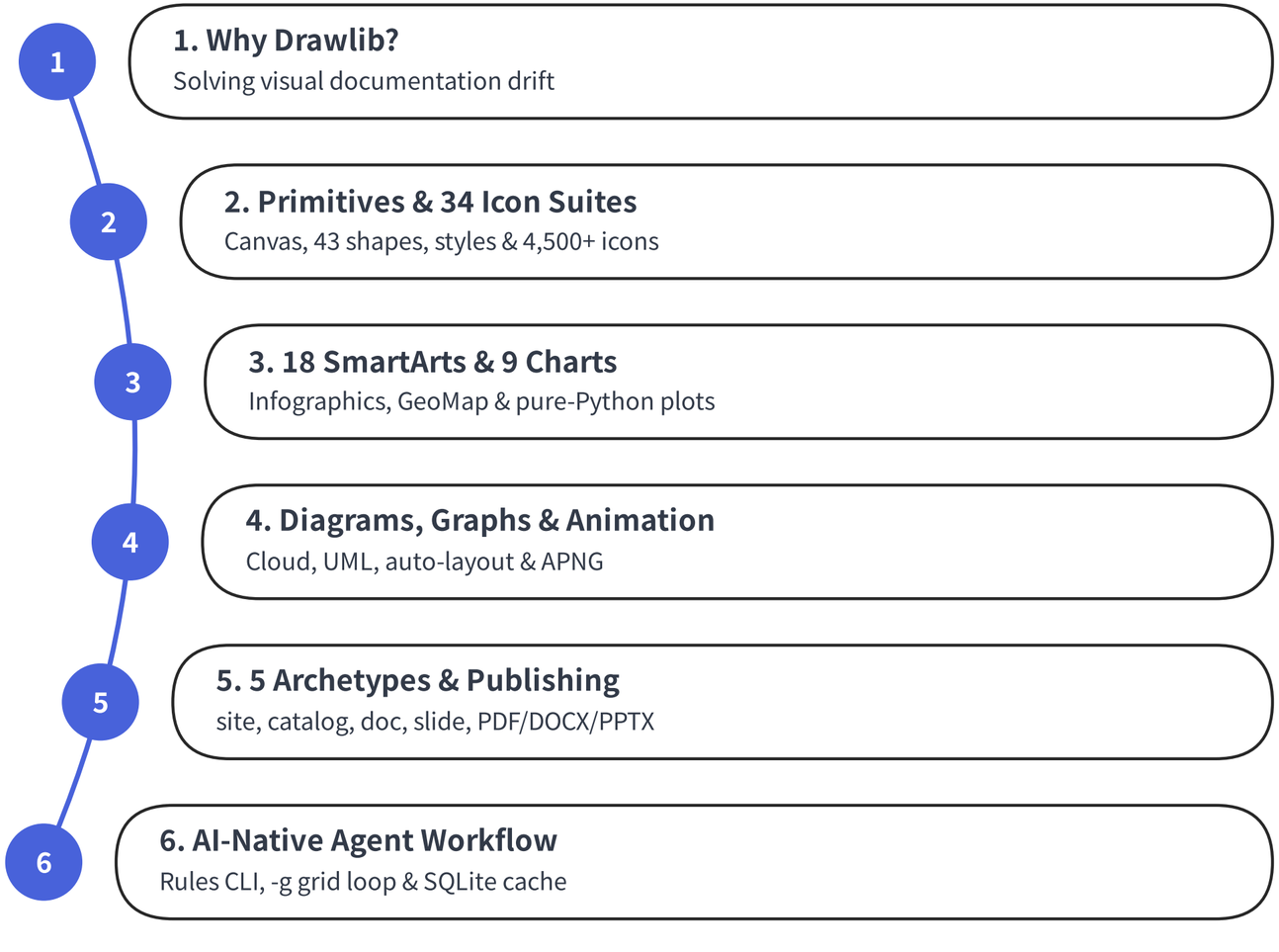
L1 · Layer 1 · AI Agent CLI

rules · show -g Grid Inspection · SQLite Cache

Presentation Roadmap

Six Core Chapters

- **Motivation (Ch. 1)**: Solving visual documentation drift with Python and Git
- **Visual Engines (Ch. 2–4)**: Primitives, 34 icon suites, SmartArts, charts, and diagrams
- **Publishing & AI (Ch. 5–6)**: Five project archetypes, Office export, and AI grid review



CHAPTER

01

Chapter 1 of 6

Why Drawlib?

Solving Visual Documentation Drift in the AI Era

1.1 Why GUI Whiteboards & Text DSLs Break Down

1.2 Illustration & Documentation as Code

1.3 Landscape Comparison Matrix

1.4 Single Source of Truth Across Formats

Why Visual Documentation Breaks Down

Engineering teams are forced to choose between **untrackable GUI binaries**, **rigid text DSLs**, and **isolated plotting libraries**—causing visual architecture docs to drift behind code on every release.

GUI Tools (Figma / Draw.io)

- **Opaque Binary Files**
  No Git PR diff review
- **Manual PNG Exports**
  Screenshots rot on release
- **GUI Lock-In**
  Uneditable by AI agents

{ } Text DSLs (Mermaid / UML)

- **Rigid Auto-Layout**
  Zero exact (x, y) control
- **Narrow Vocabulary**
  Weak KPIs, tables & maps
- **Snippet-Only Scope**
  No PDF or PPTX compiler

Plotting Libs (Matplotlib)

- **Data Plots Only**
  No cloud or UML engines
- **Verbose Boilerplate**
  Ad-hoc fonts and colors
- **Detached Scripts**
  Manual copy into slides

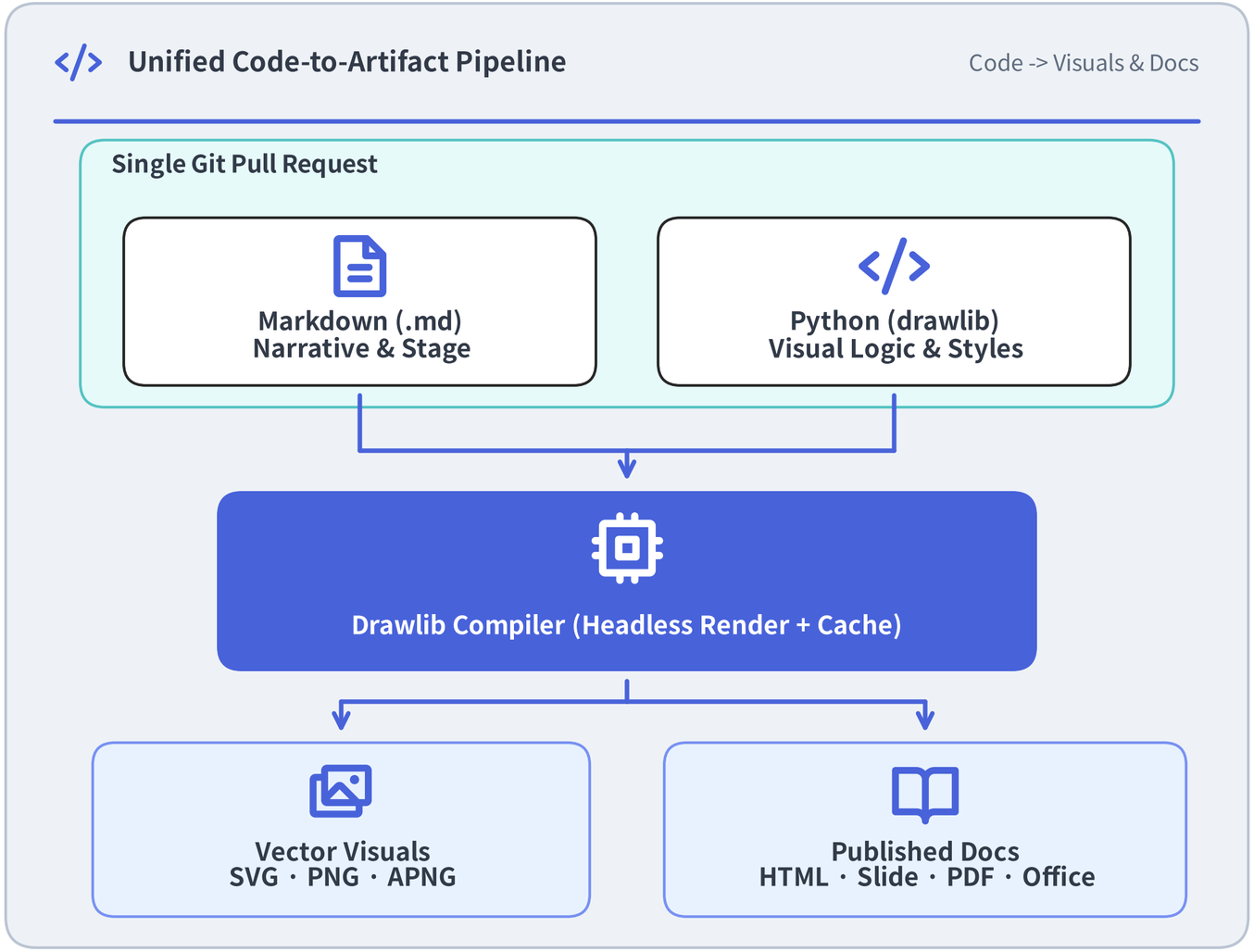
The Unmet Gap: Spatial Coordinates + Git PR Review + Unified Docs & Charts + AI Grid Verification

Illustration & Documentation as Code

Pure Python + Markdown

- **Turing-Complete Layout**: Loops, variables, and shared `utils.py` helpers
- **Inline Code Blocks**: Embed `drawlib` visuals directly inside Markdown
- **Deterministic PR Diffs**: Review prose and diagram updates together

```
setup(width=120, height=32)
p = ChevronProcess(style=Styles.Neutral)
for s in ("Code", "Build", "Ship"): p.add(s)
p.draw((6, 8), width=108, height=16)
save()
```

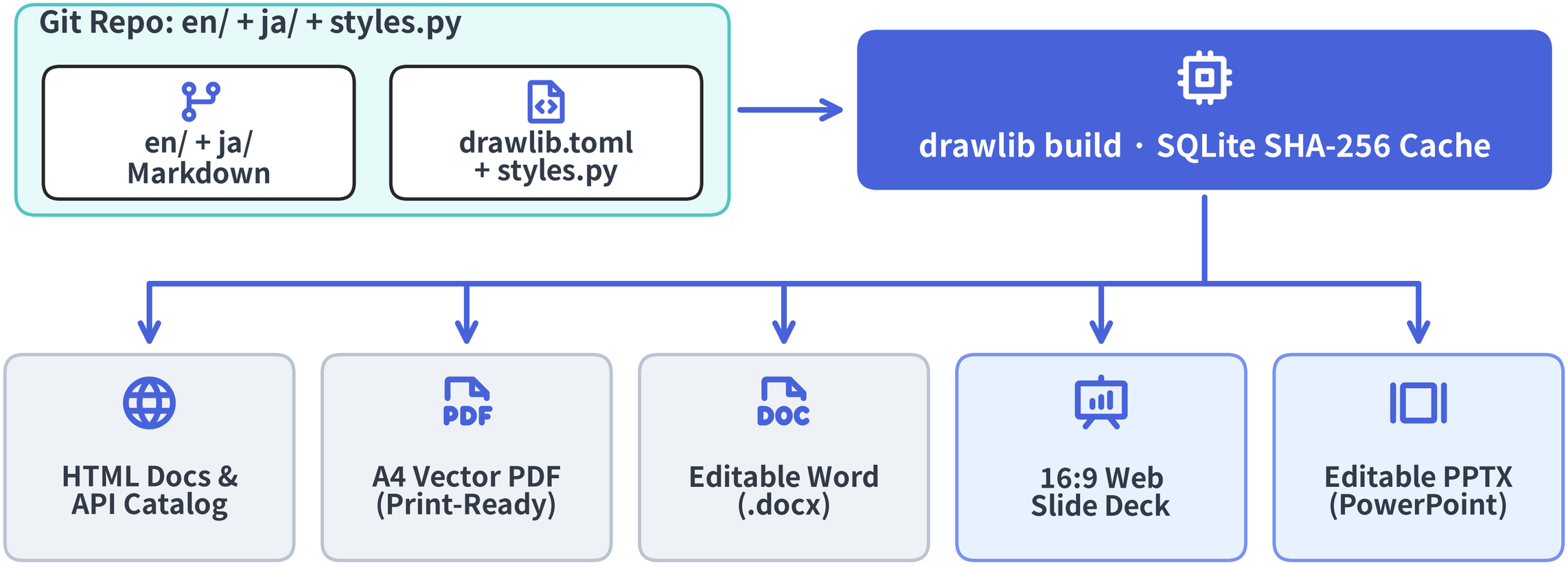


Tooling Landscape Comparison

Capability	GUI Whiteboards	Text DSLs	Plotting Libs	Drawlib
Git Diff & PR Review	× Binary Blob	✓ Text Syntax	✓ Python Code	**✓ Clean Python**
Spatial Coordinate Control	✓ Manual Drag	× Rigid Auto	△ Axes Only	**✓ Exact + Auto**
Cloud & UML Diagrams	✓ Manual Icons	△ Basic Boxes	× Unsupported	**✓ 6 Engines + Icons**
Charts & Infographics	△ Manual Shapes	× Very Limited	✓ Plots Only	**✓ Charts & SmartArts**
Site / Catalog / Slide / Office	× Image Export	× Snippet Only	× Image Only	**✓ 5 Archetypes**
AI Visual Self-Healing (-g)	× GUI Locked	× Blind Text	△ Ad-Hoc Scripts	**✓ Built-In -g CLI**

One Git Repository, Every Output Format

A single `drawlib.toml` manifest with `[i18n]` structural parity compiles synchronized **English and Japanese** deliverables across Web, Print, and Office from one Git commit.



CHAPTER

02

Chapter 2 of 6

Primitives, Styles & Icons

43 Shapes, Semantic Themes & 4,500+ Vector Icons

2.1 Cartesian Canvas & -g Visual Grid

2.2 43 Shapes & Smart Connectors

2.3 Multilingual Typography & Fonts

2.4 34 Icon Suites & 8-Color Themes

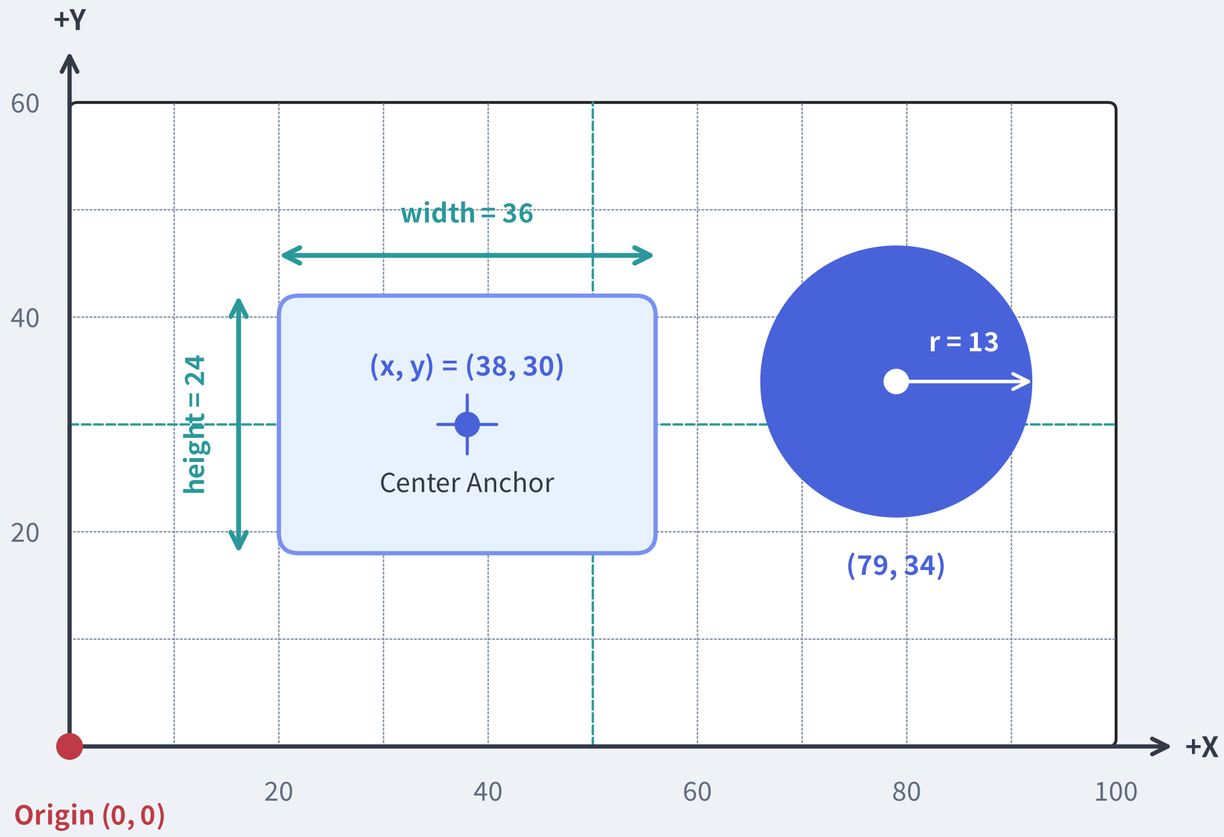
Cartesian Canvas & Visual Grid (-g)

Predictable Math Geometry

- **Bottom-Left Origin**: `(0, 0)` starts bottom-left (`+X` right, `+Y` up).
- **Center Anchor**: `(x, y)` centers every shape, text, and icon.
- **Visual Grid (`-g`)**: Overlay coordinate ticks for exact verification.

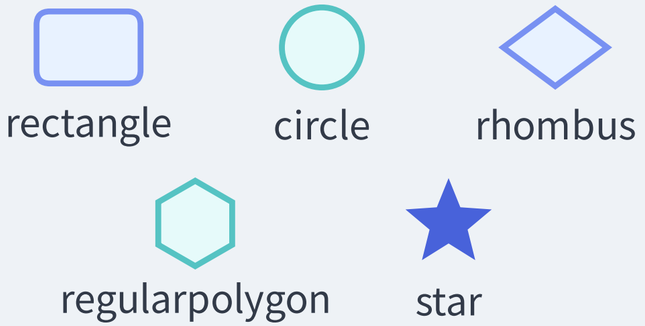
```
setup(width=100, height=60)
st = Styles.PrimaryFlat
rectangle((50, 30), 36, 22, style=st)
save()
```

Cartesian Stage: setup(width=100, height=60) + Grid (-g)



43 Geometric Shapes & Smart Connectors

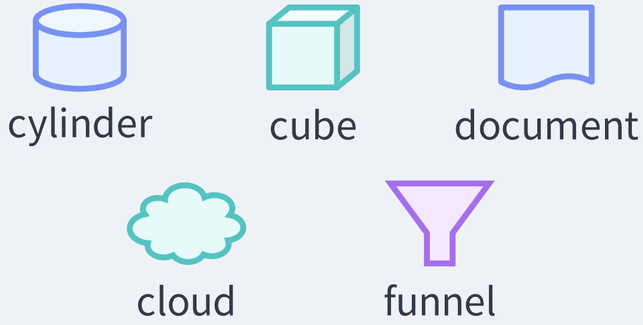
1. Basic & Polygons



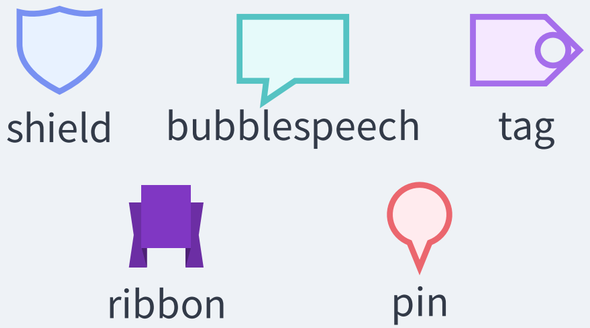
2. Block Arrows & Flow



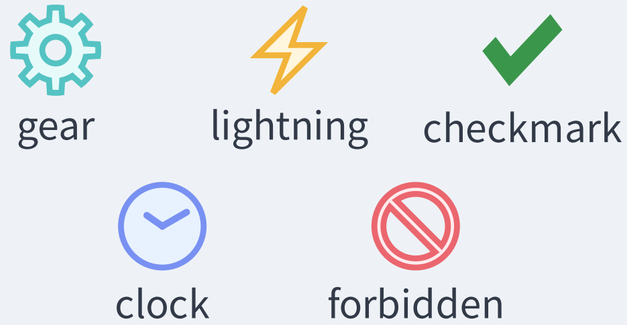
3. 3D & Containers



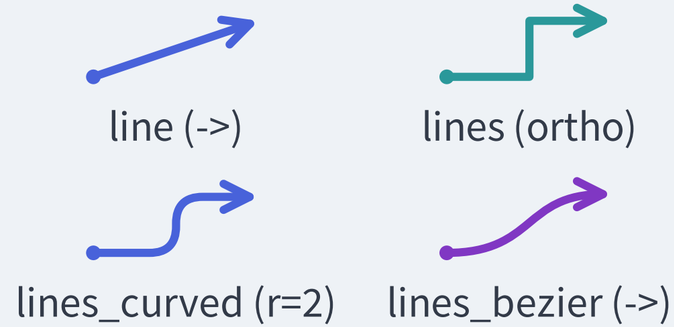
4. Badges & Callouts



5. Domain Symbols



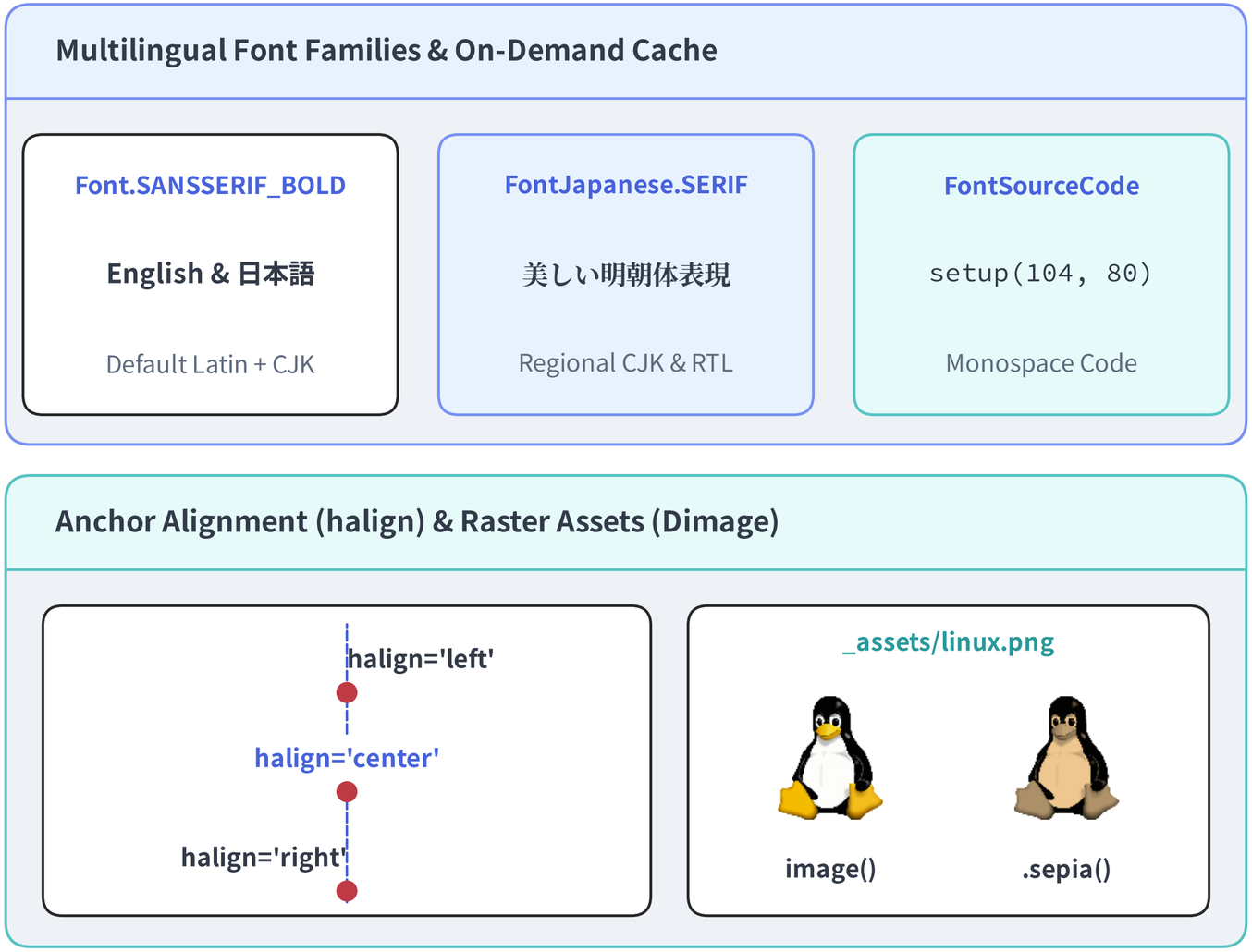
6. Lines & Curves



Multilingual Typography & Raster Assets

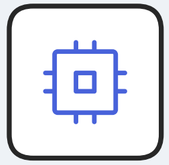
Universal Scripts & Dimage

- **CJK + Latin Default**: `Font` renders English and Japanese together.
- **Regional Packs**: `FontJapanese`, `FontArabic`, and `FontSourceCode`.
- **Exact Text Anchors**: Control alignment via `halign` and `valign`.
- **Raster** `Dimage`: Crop, resize, sepia, mosaic, plus font caching.



34 Icon Suites & 4,500+ Vector Icons

1. Phosphor Vector Font Icons (5 Weights)

thin	light	regular	bold	fill
				
rocket_launch	cpu	globe	sparkle	shield_check

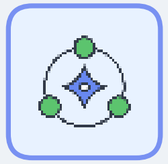
2. Official Cloud Suites (gcp · aws · azure)

gcp	gcp	aws	aws	azure
				
Cloud Run	BigQuery	Lambda	Amazon S3	Cosmos DB

3. 15 Official Domain Brand Suites

ai	cnative	devops	data	security
				
Gemini	Kubernetes	GitHub	PostgreSQL	Cloudflare

4. 15 Original 2D Suites (original_*)

cloud	ai	server	network	security
				
multi_cloud	ai_agent	gpu_server	firewall	zero_trust

8-Color Semantic Styles & Preset Themes

- **Semantic Matrix**: `Styles.<Color><Variant>` combines 8 roles (`Primary` .. `Dark`) with `Flat` , `Bold` , `Dashed` , and `Neutral` .
- **Immutable `.patch()` & Themes**: Customize safely via `style.patch(...)` or switch global themes in `drawlib.toml` .

8 Semantic Roles × Variants (Styles.<Color><Variant>)

Role	Default	Flat	Neutral / Dashed
Primary	.Primary	.PrimaryFlat	.PrimaryNeutral
Secondary	.Secondary	.SecondaryFlat	.SecondaryNeutral
Accent	.Accent	.AccentFlat	.AccentNeutral
Success	.Success	.SuccessFlat	.SuccessNeutral
Warning	.Warning	.WarningFlat	.WarningNeutral
Danger	.Danger	.DangerFlat	.DangerNeutral
Neutral	.Neutral	.NeutralFlat	.NeutralDashed
Dark	.Dark	.DarkFlat	.DarkDashed

Preset Themes (drawlib.toml)

style = "default"
Indigo Editorial Palette



style = "google"
Google Workspace Blue



style = "monochrome"
Print & Paper Grayscale



CHAPTER

03

Chapter 3 of 6

SmartArts & Pure-Python Charts

18 Structured Infographics & 9 Zero-Dependency Plots

3.1 Process Flows, Cycles & Roadmaps

3.2 KPIs, CardLists, Pyramids & MindMaps

3.3 Tables, SourceCode, Calendars & GeoMap

3.4 9 Declarative Chart Types & Gantt

Process Pipelines, Lifecycles & Roadmaps

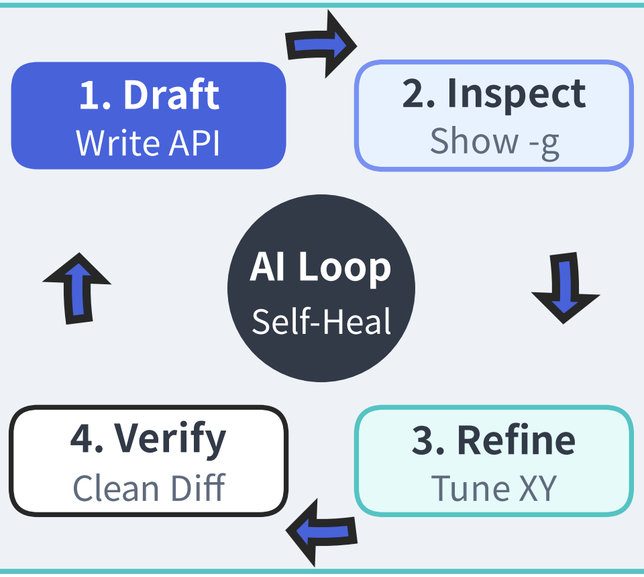
ChevronProcess — Sequential Delivery Pipeline

4 Stages

1. **Author** — Markdown + Python
2. **Render -g** — Visual Grid Check
3. **Cache** — SQLite Incremental
4. **Publish** — HTML, PDF & PPTX

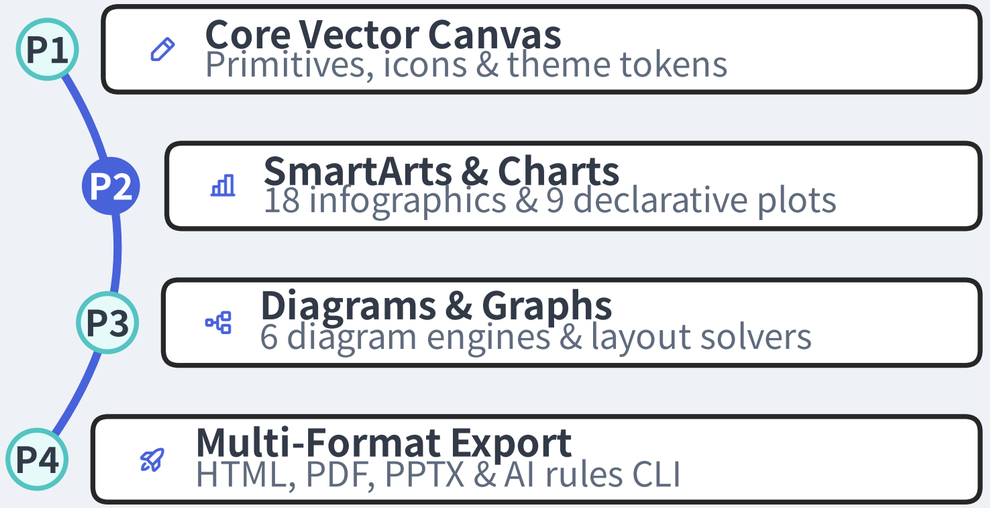
Cycle — Feedback Loop

4-Node Orbit



Roadmap — Milestone Timeline

Bezier Spine



KPIs, Feature Cards, Pyramids & MindMaps

KpiCards — Metric Grid

2x2 Layout

18	**SmartArt Types** Tables to GeoMaps
34	**Icon Suites** Cloud & Phosphor
0	**GUI Lock-In** 100% Git Python
4	**Theme Palettes** Instant re-skin

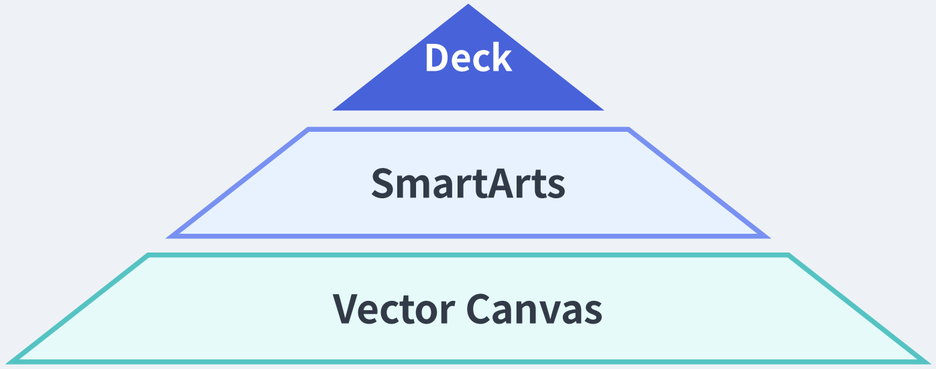
CardList — Feature Stack

Icon + Badge

- L1 **Declarative Item Lifecycle** Register via .add() and chain .draw()
- L2 **Semantic Design Tokens** 50%+ neutral base with focal highlights
- L3 **Animation-Ready Visibility** Toggle item.show without shifting layout

Pyramid — Layered Hierarchy

3 Tiers



MindMapNode — Radial Tree

2-Pass Layout



Tables, SourceCode, Calendars & GeoMap

Table — Benchmark Matrix

Zebra Rows

Output Target	Format	Cold Build
16:9 Slide Deck	HTML / PDF	**0.42 s**
Documentation Site	Multi-Page	**0.65 s**
Standalone Diagram	SVG / PNG	**0.18 s**

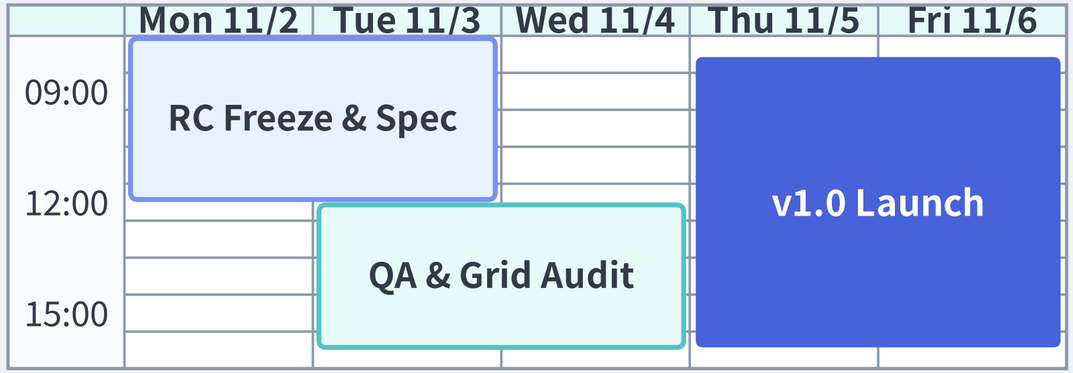
SourceCode — Syntax Box

card_demo.py

```python
from drawlib.canvas import save, setup
from drawlib.shapes import rectangle
from drawlib.styles import Styles
setup(width=120, height=60)
rectangle((60, 30), 80, 36, Styles.Primary)
save("card.svg")
```

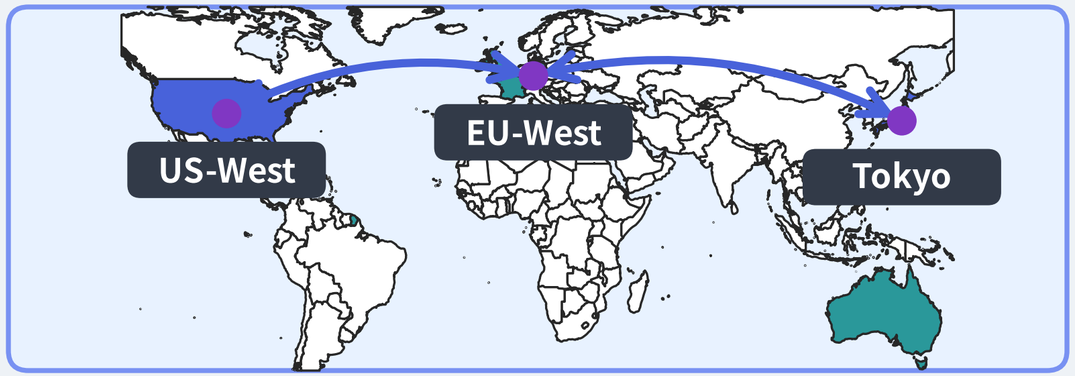
WeekCalendar — Sprint Plan

ISO 8601



GeoMap — Global Regions

US, EU & JP



Declarative Bar, Line & Area Charts

9

Built-In Chart Types

Bar, Line, Area, Pie,
Radar, Scatter, Gantt, Heatmap

100%

Theme-Synchronized

Matches Default, Google &
Monochrome palettes

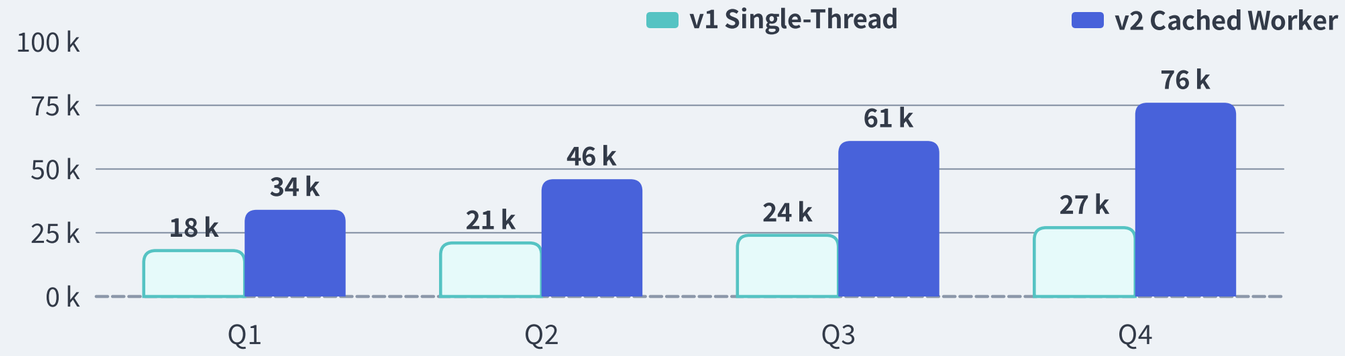
Auto

Nice-Number Axes

Clean ticks, legends &
progressive draw_ratio

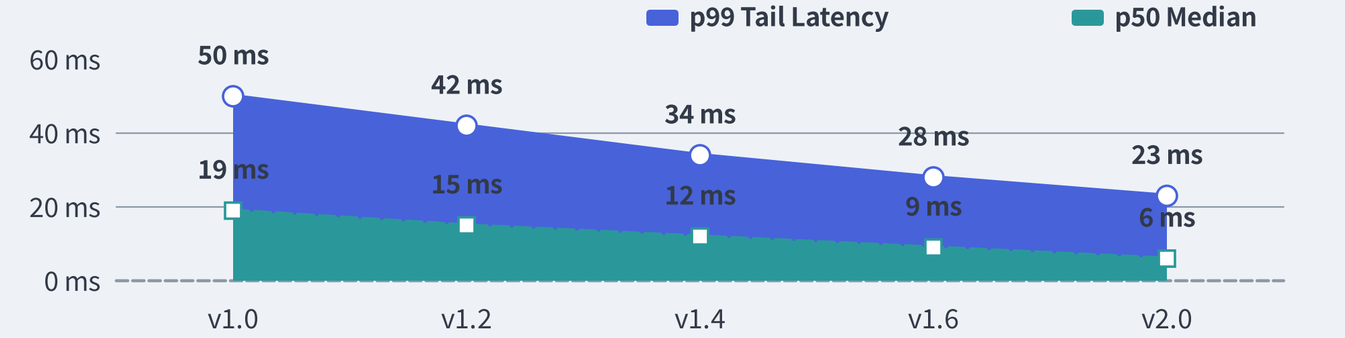
BarChart — Quarterly Render Throughput (k/hr)

Grouped Mode



AreaChart — Build Pipeline Latency (p50 vs p99)

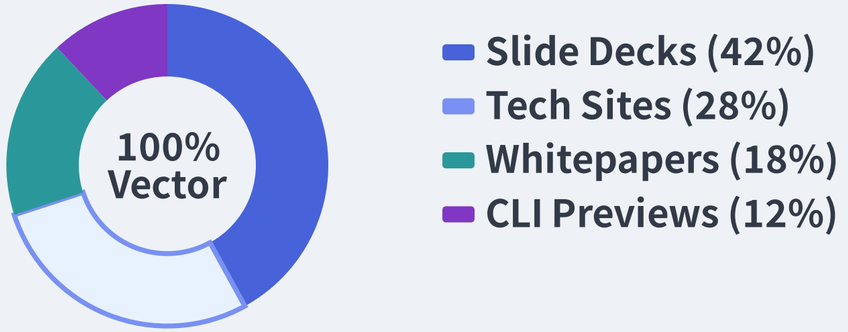
Smooth Spline



Pie, Radar, Scatter & Heatmap Charts

PieChart — Donut Share

hole_ratio=0.55



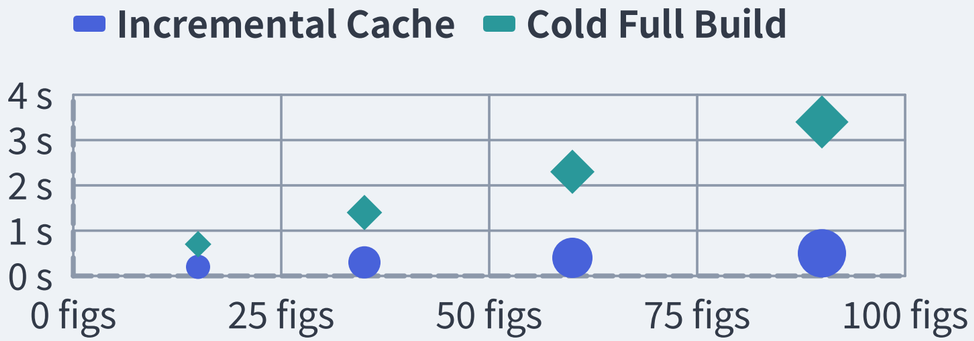
RadarChart — 5-Axis Profile

Polygon Web



ScatterChart — Build Scaling

Bubble Series



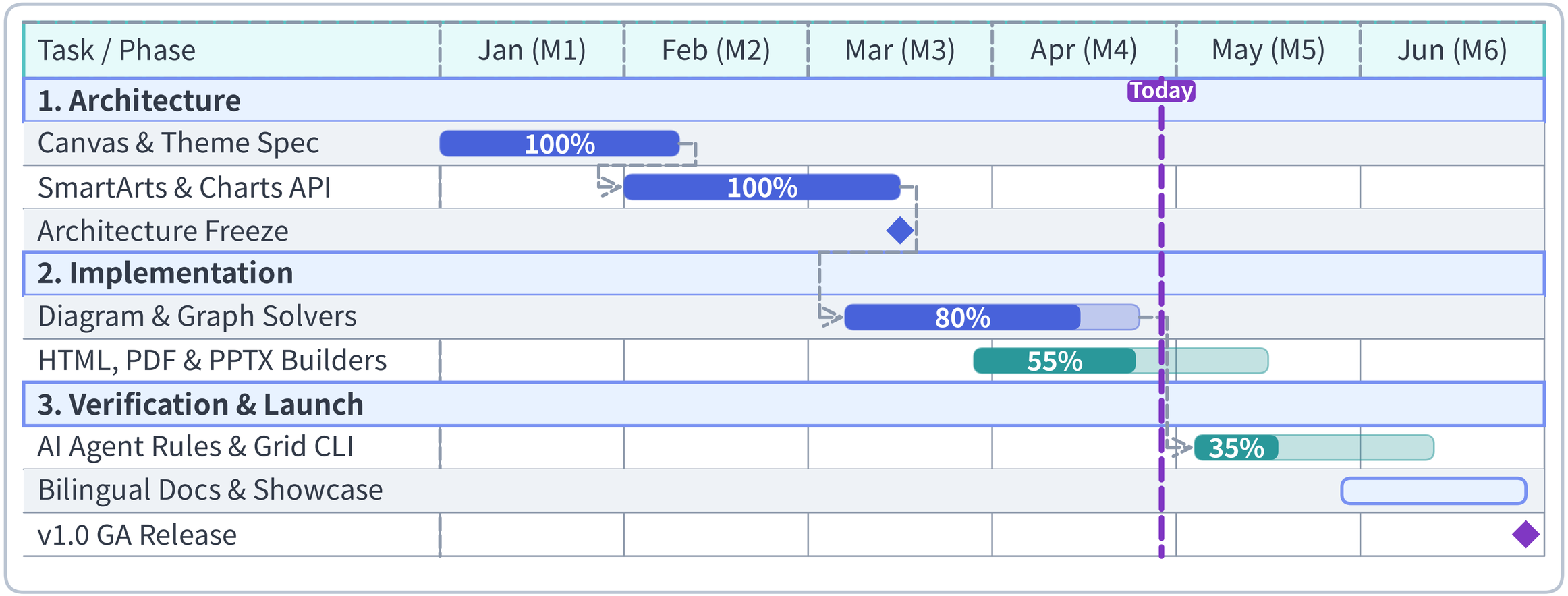
HeatmapChart — Latency

Auto-Contrast

	00:00	06:00	12:00	18:00	22:00
US-West	14ms	18ms	42ms	35ms	19ms
EU-West	16ms	29ms	54ms	38ms	21ms
Tokyo	22ms	46ms	68ms	49ms	25ms
Singapore	18ms	34ms	51ms	41ms	22ms

Engineering Roadmaps with GanttChart

- **Declarative Schedule Modeling**: Group workstreams with `Section`, track completion via `Task(progress=...)`, and pin `Milestone` diamonds.
- **Dependencies & Live Status Markers**: Route orthogonal `Dependency` arrows and overlay vertical `Marker` lines (`Today`, `Release`) in code.



CHAPTER

04

Chapter 4 of 6

Diagrams, Graphs & Animations

6 Engineering Diagram Engines, 5 Solvers & APNG

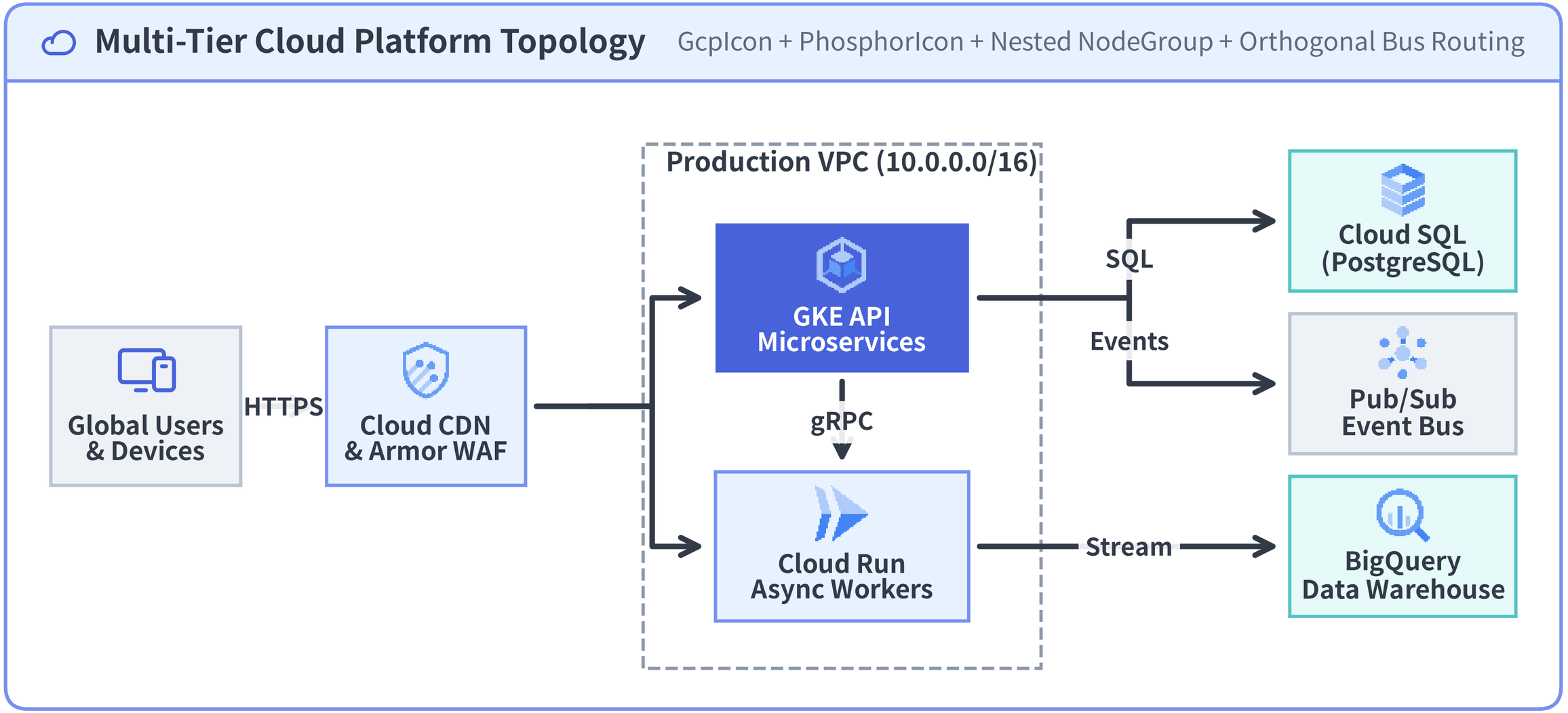
4.1 Cloud Topology (ArchitectureDiagram)

4.2 Flowcharts & API Sequence Diagrams

4.3 UML Class, ER & State Machines

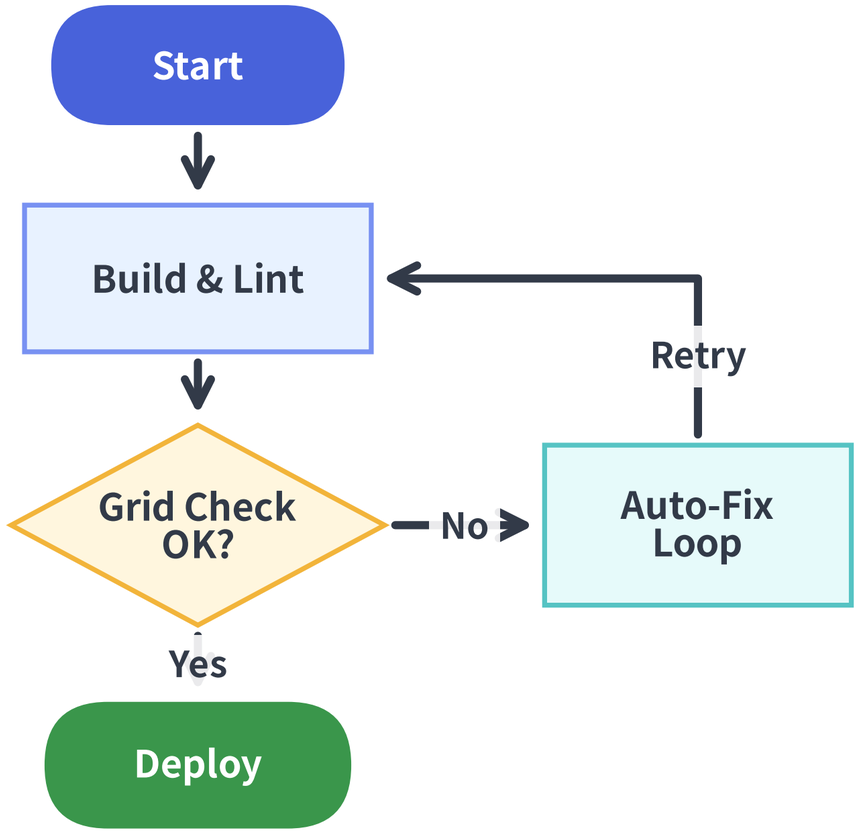
4.4 Auto-Layout Graphs & Animations

Cloud Topology with ArchitectureDiagram

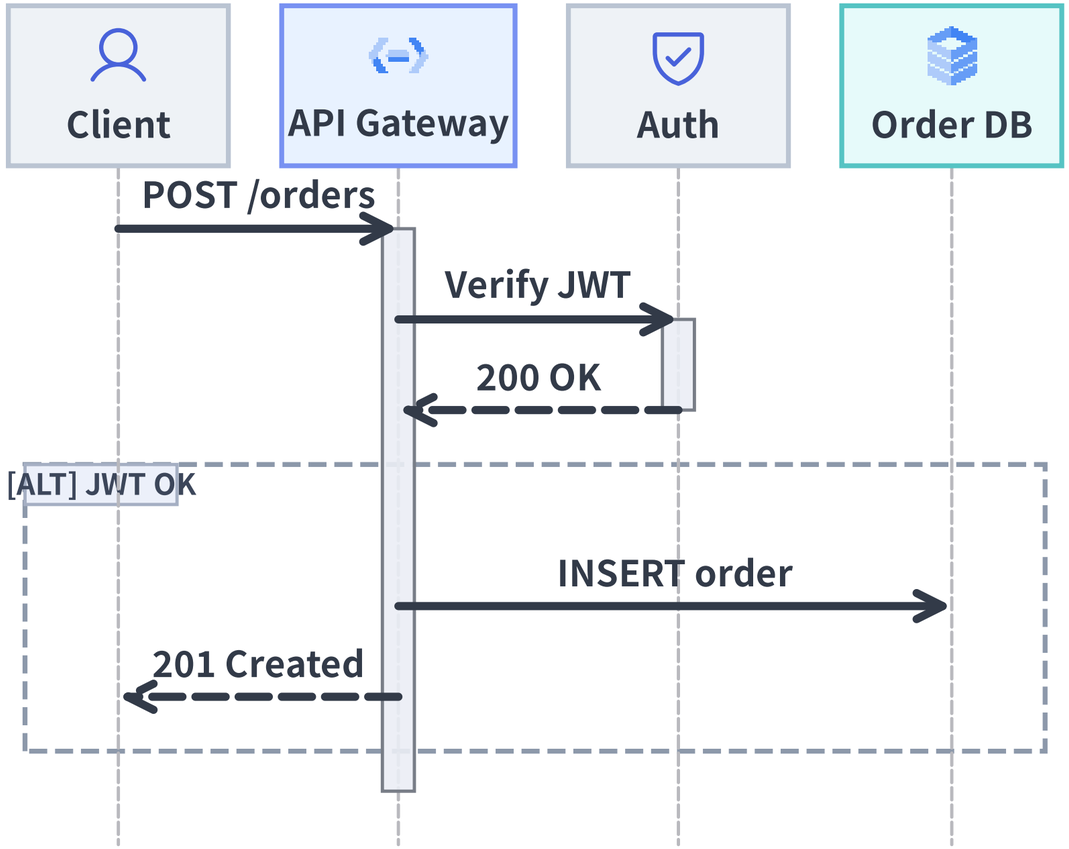


Workflows & Protocols: Flow and Sequence

FlowDiagram — CI/CD Validation

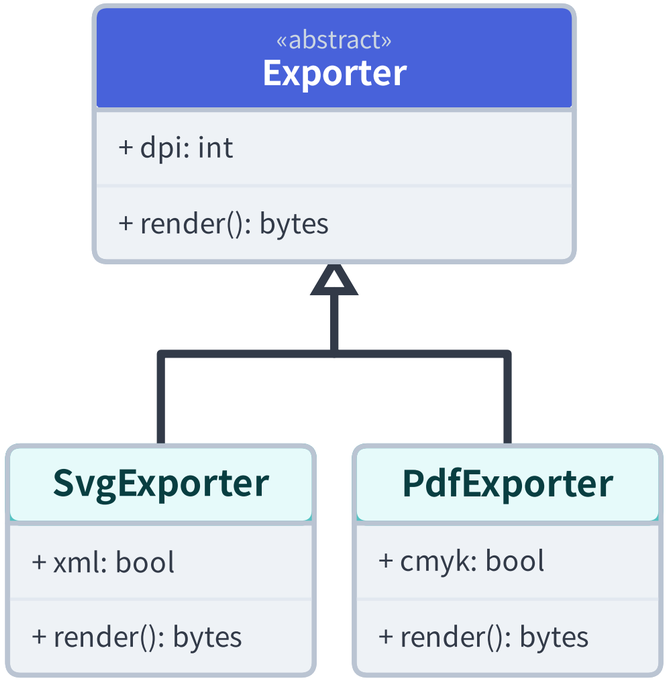


SequenceDiagram — API Protocol



Domain Modeling: Class, ER & State Diagrams

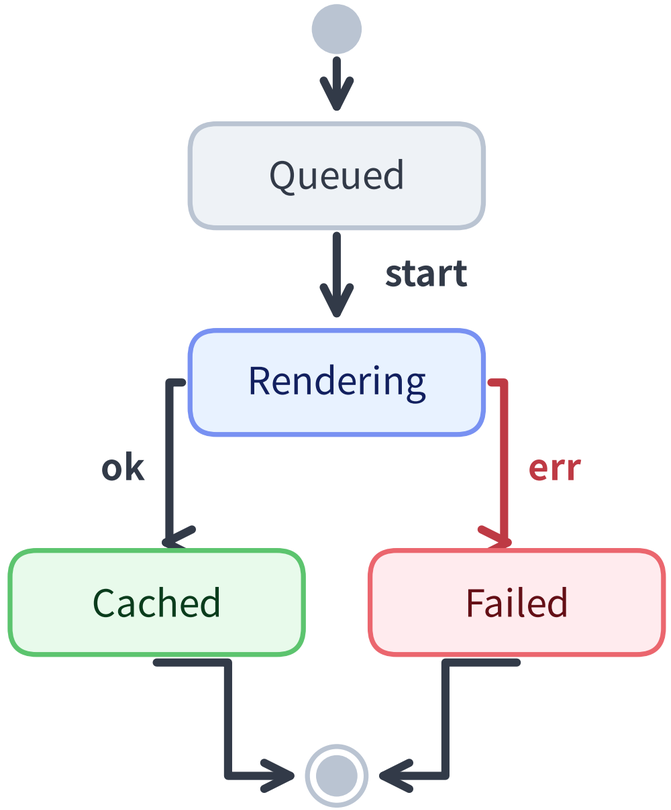
ClassDiagram (UML 2.0)



ERDiagram (Schema)



StateDiagram (FSM)

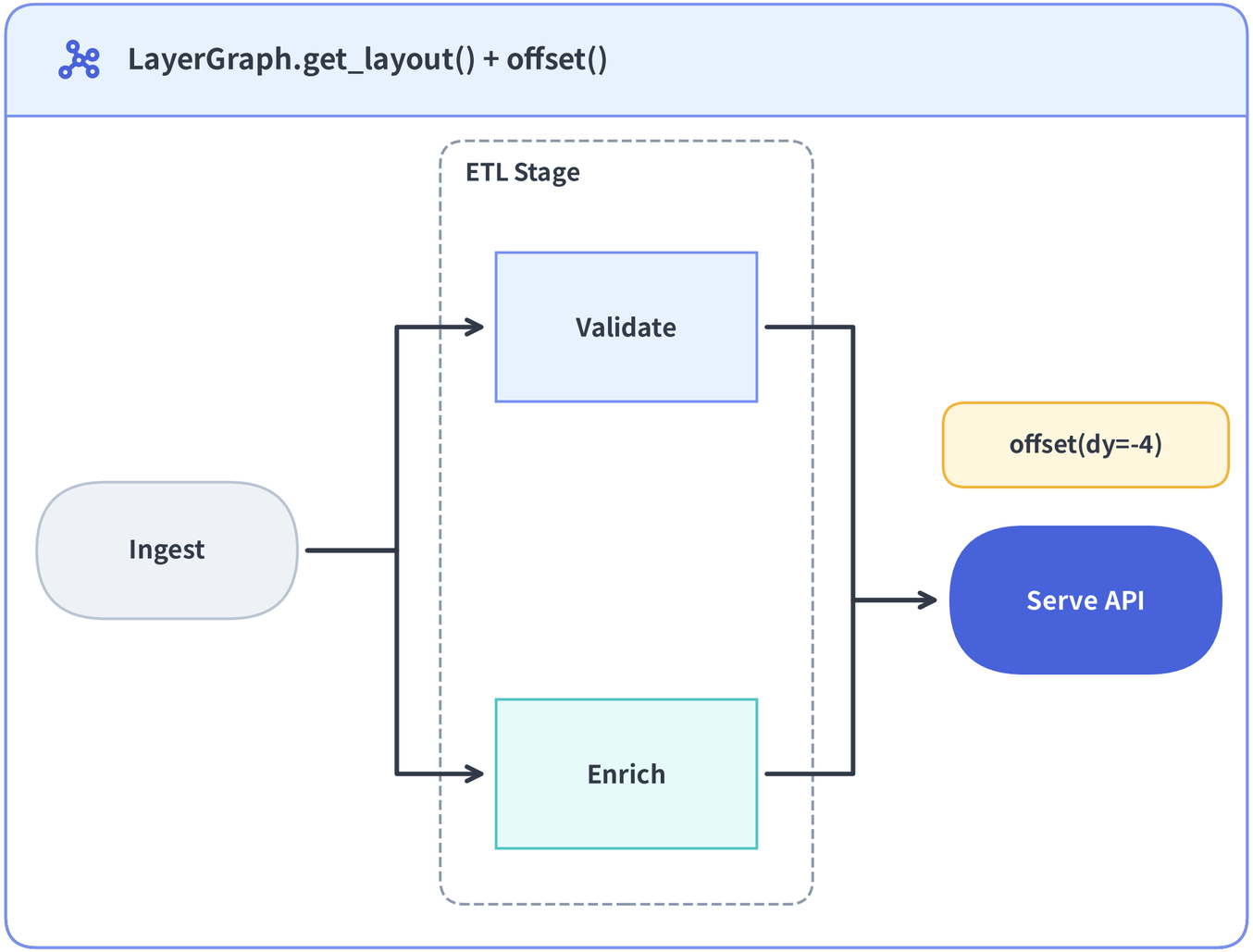


Declarative Graph Layout (drawlib.graph)

Hybrid Algorithmic + Manual Control

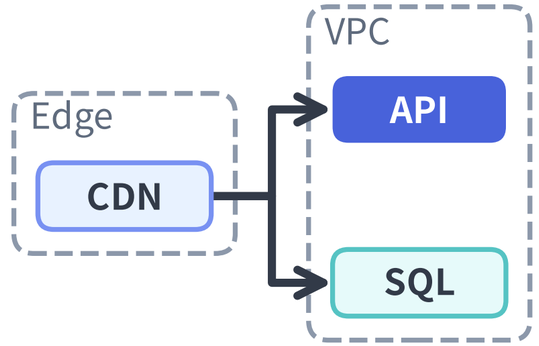
- **Auto-Solve Topology**: Crossing-free `(x, y)` via `get_layout()`
- **Manual Fine-Tuning**: Nudge nodes with `offset(dx, dy)` anytime
- **No Black-Box Lock-In**: Inspect & freeze code via `print(layout)`

```
g = LayerGraph(direction="LR")
g.edge("ingest", "validate")
g.edge("validate", "serve")
layout = g.get_layout(margin=8.0)
layout.offset("serve", dy=-4.0)
```

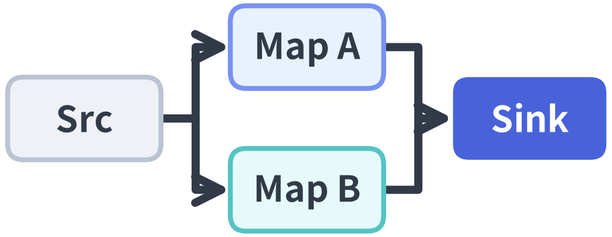


Five Specialized Auto-Layout Graph Solvers

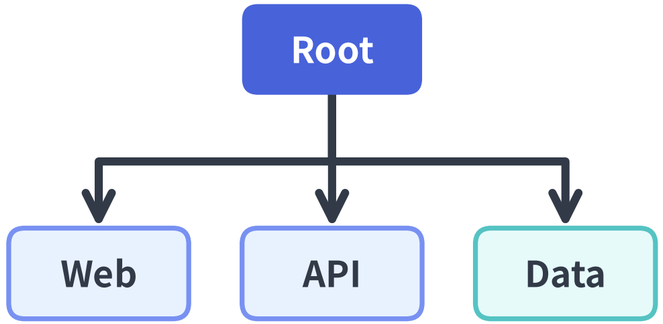
1. ArchitectureGraph



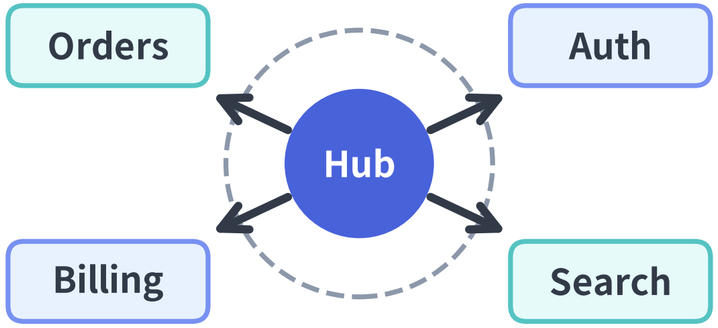
2. LayerGraph (DAG Tiers)



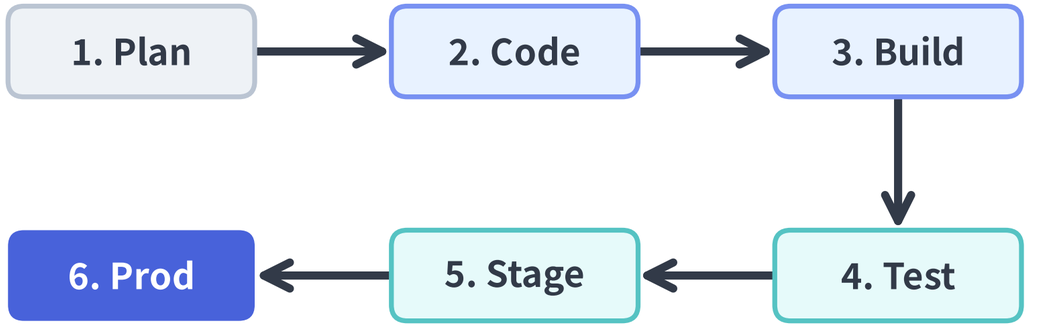
3. TreeGraph (Tidy Tree)



4. RadialGraph (Hub & Concentric Spokes)



5. GridGraph (2D Matrix & Snake Flow)

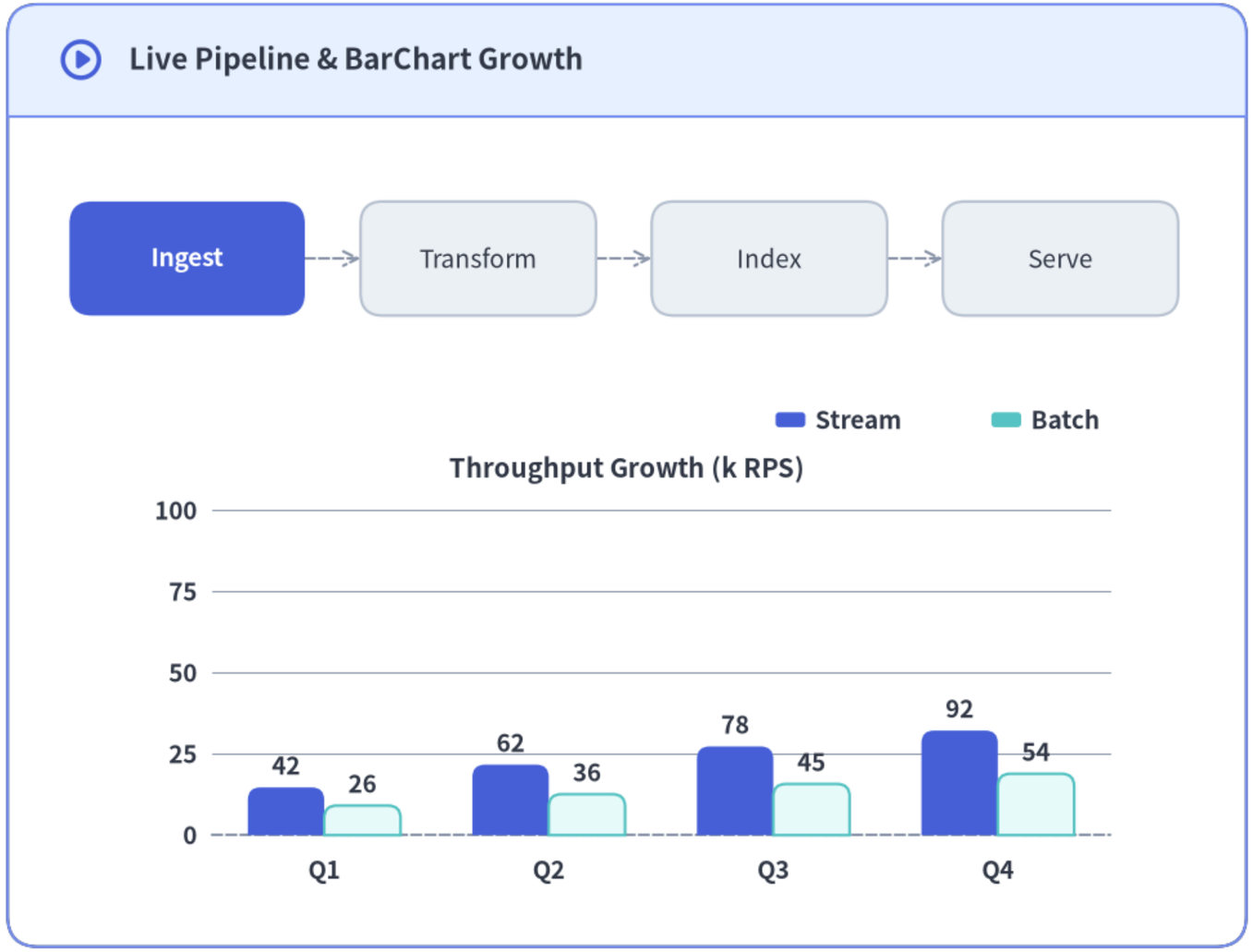


Interactive Frame Animations (APNG & WebP)

Motion as Code Across All Modules

- **Multi-Frame Engine**: Declarative `Animation(fps, loop)` API
- **Progressive Reveal**: `draw_ratio` on Lines, Charts & Diagrams
- **Interactive Stage**: Live `<canvas>` (`anim_trigger` , `anim_pause` , `A`)
- **Static Export Ready**: Auto-extracts clean frames for PDF & PPTX

```
anim = Animation(fps=2.0, loop=0)
for r in [0.4, 0.7, 1.0]:
    series.draw_ratio = r
    with anim.frame():
        chart.draw(xy=(10, 6))
```



CHAPTER

05

Chapter 5 of 6

5 Archetypes & Publishing

Site, Catalog, Doc, Slide & Images from One Compiler

5.1 Five Project Archetypes Overview

5.2 Web Docs (site) & API Reference (catalog)

5.3 Whitepapers: HTML, Vector PDF & DOCX

5.4 16:9 Slide Stage, Presenter View & PPTX

Five Built-In Project Archetypes

`uv run drawlib init project <type> <dir>` — One Compiler • 5 Purpose-Built Scaffolds • Web, PDF & Office

site

`docs_src/<lang>/`

Guided narrative docs
navbar.md + search
Dark mode & i18n

e.g. www.drawlib.com

HTML | Search | SVG

catalog

NEW in v0.3.5

`catalog_src/<lang>/`

Two-column API ref
Auto # H1 / ## H2
Instant / filter

e.g. drawlib.com/api

HTML | / Filter | SVG

doc

`doc_src/<lang>/`

Linear whitepapers
--generate-index
Auto Cover & ToC

RFCs & Design Specs

HTML | PDF | DOCX

slide

`slide_src/<lang>/`

16:9 visual stage
?presenter=1 sync
::: note scripts

Keynotes & Tech Talks

HTML | PDF | PPTX

images

`images_src/*.py`

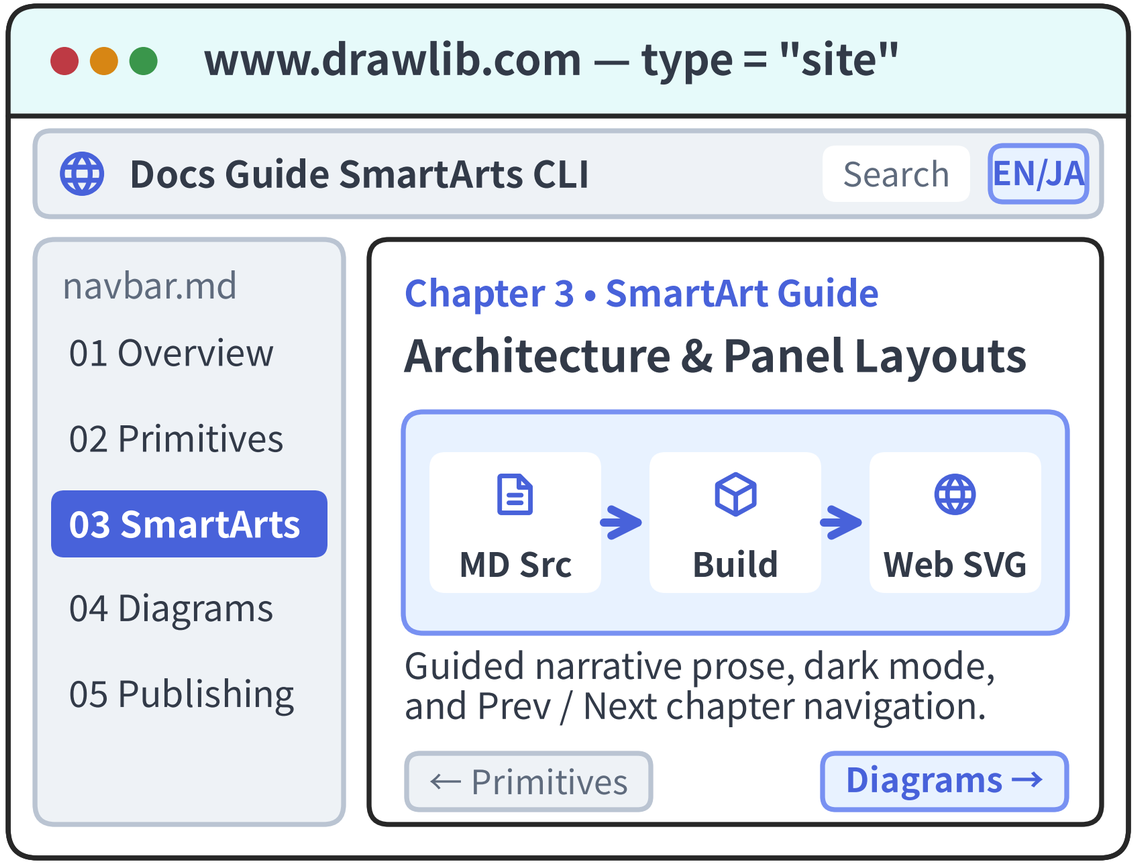
Standalone Python
Batch illustration
CI asset generation

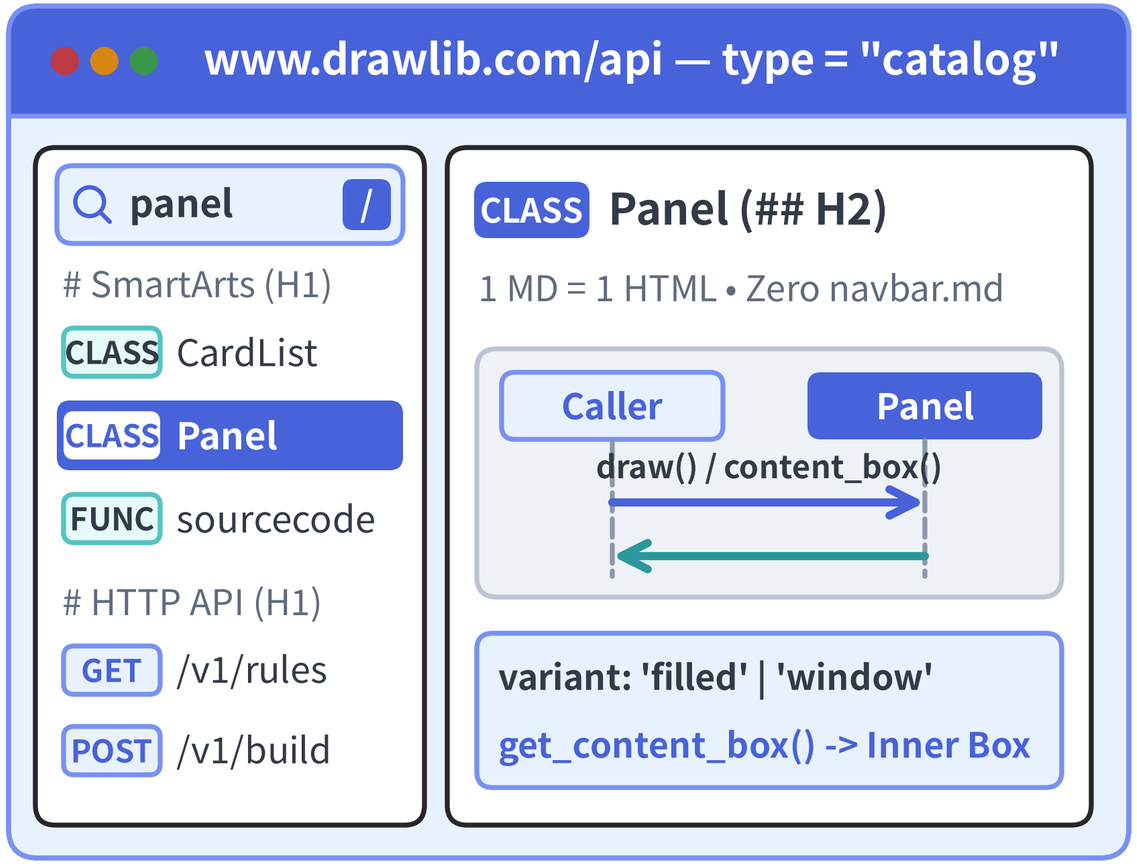
README & Paper Art

SVG | PNG | WebP

Web Publishing: Docs Site & API Catalog

- `type = "site"` **(Narrative Docs)**: Guided chapter navigation via `navbar.md`, full-text search, dark mode & `i18n` (`www.drawlib.com`).
- `type = "catalog"` **(API Reference)**: Zero-`navbar.md` `1 MD = 1 HTML` dictionary extracting `# H1` / badged `## H2` items with `/` instant filter (`www.drawlib.com/api`).



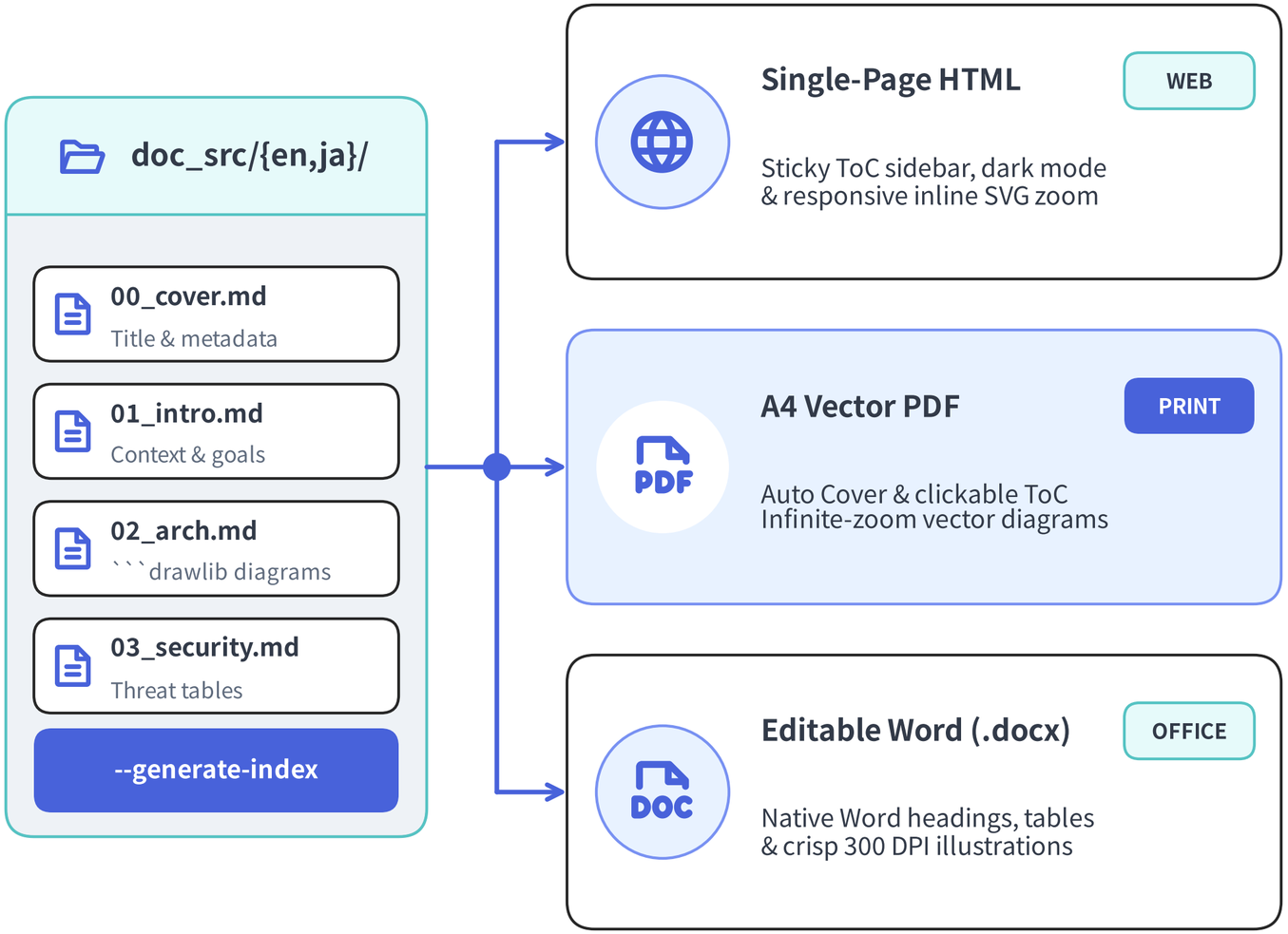


Whitepapers: HTML, Vector PDF & MS Word

Linear Specs (`type = "doc"`)

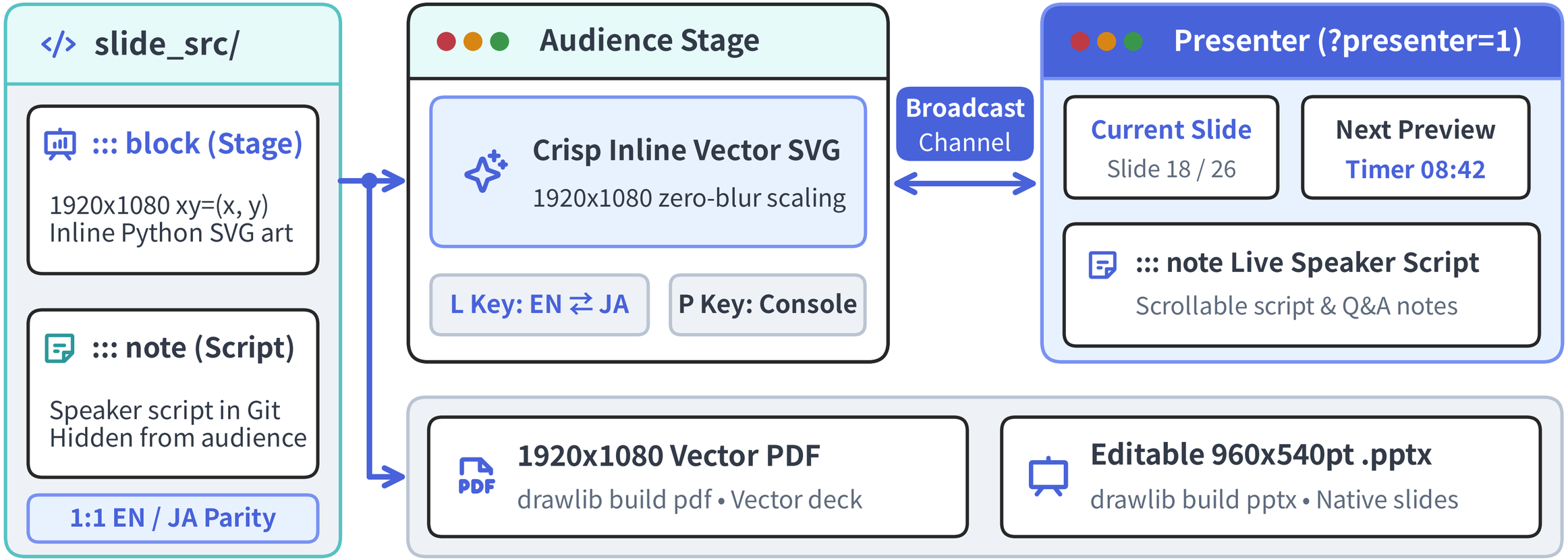
- **Numbered Chapters**: Merge `00_cover.md`, `01_intro.md` into one spec
- **Auto Cover & ToC**: `--generate-index` builds cover & clickable index
- **Triple Export**: Compile HTML, A4 Vector PDF, and editable `.docx`

```
# Build all 3 deliverables from doc_src/
uv run drawlib build html
uv run drawlib build pdf --generate-index
uv run drawlib build docx --generate-index
```



16:9 Slide Stage, Presenter View & PPTX

- **Dual-Layer Slide Markdown**: `::: block` places crisp visuals on a `1920×1080` stage while `::: note` stores speaker scripts in Git.
- **Live Sync & Office Export**: Sync Audience Stage with `?presenter=1` Presenter View via `BroadcastChannel`, plus Vector PDF & `.pptx`.



CHAPTER

06

Chapter 6 of 6

AI-Native Agent Workflow

Rules CLI, Visual Self-Correction & SQLite Caching

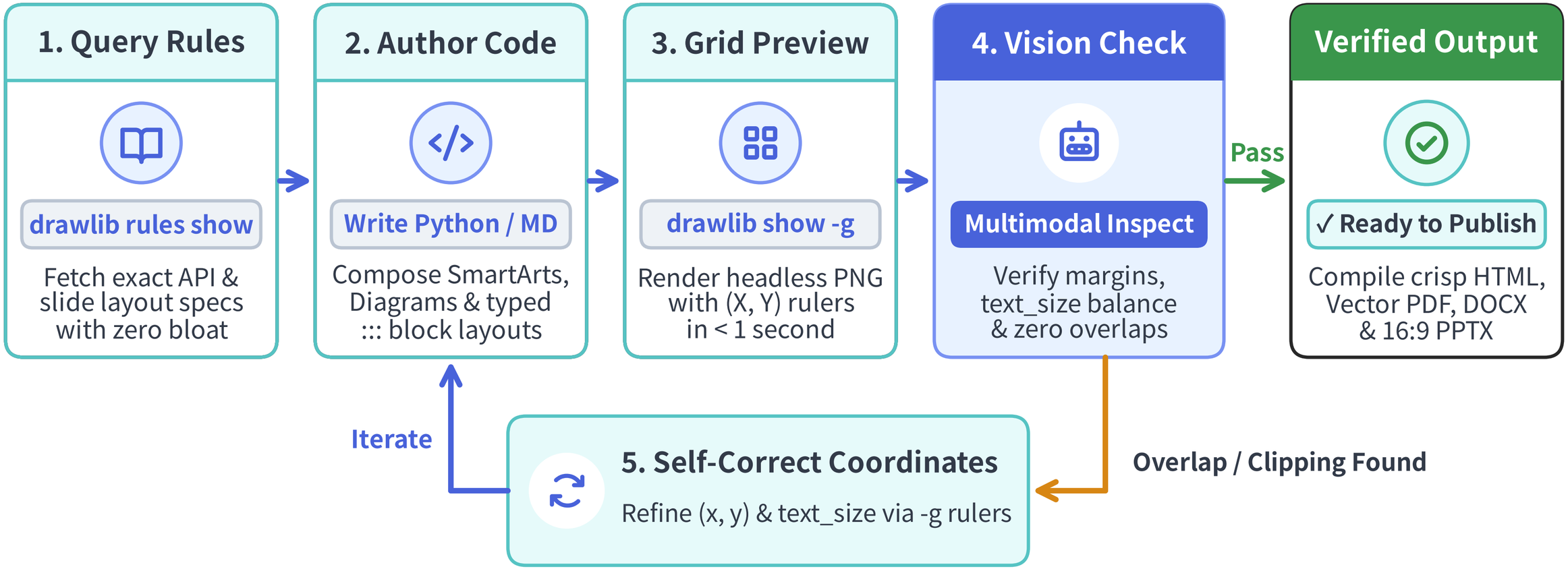
6.1 On-Demand Context via drawlib rules

6.2 Autonomous -g Grid Verification Loop

6.3 Two-Tier Asset & SQLite Build Cache

Autonomous Multimodal Verification Loop

- **On-Demand Context**: AI agents (Gemini, Claude Code, Cursor) query exact API specs via `uv run drawlib rules show <topic>`.
- **Visual Self-Correction**: Agents inspect `-g` coordinate grid previews multimodally and fix overlaps before publishing.



Two-Tier Caching & Unified CLI

< 1s

Incremental Builds

SQLite SHA-256 .drawlib/cache.db

50+

Agent Rule Topics

On-demand via drawlib rules show

Zero

Broken Links

Built-in drawlib serve --check

Two-Tier Cache Architecture

Tier 1: Asset Cache

drawlib cache list / download

On-demand fonts, 34 icon packs
& headless Chromium binary

Tier 2: Build Cache

.drawlib/cache.db (SQLite)

SHA-256 content + style hash
Shared by HTML/PDF/Office

Unified CLI — uv run drawlib <command>



CLOSING 06 • Wrap-Up

Summary & Next Steps

Start Building Illustrated Documentation as Code

Key Takeaways & Quickstart

01

1. Pure Python & Git Native

Typed Python API, styles.py tokens & clean Git PR diffs — zero GUI lock-in.

02

2. Rich High-Level Components

18 SmartArts, 9 Charts, 6 Diagrams, 5 Graph solvers & 34 Icon Packs built in.

03

3. 5 Project Archetypes & Office Export

Compile Site, Catalog, Doc, Slide & Images to HTML, Vector PDF, DOCX & PPTX.

04

4. Built for Autonomous AI Agents

50+ drawlib rules topics, show -g grid QA & < 1s SQLite .drawlib/cache.db builds.

QUICKSTART IN 30 SECONDS

Requires Python 3.12+ & uv

```
uv add drawlib → uv run drawlib init project slide my-deck → uv run drawlib serve
```

Thank You

Start Authoring Visuals & Docs as Code

- **Docs**: `https://www.drawlib.com/`
- **API Catalog**: `https://www.drawlib.com/api/`
- **Examples & GitHub**: `https://examples.drawlib.com/` · `github.com/yuichi110/drawlib`

